Resonant Sensors, with Controls

What a month of resonance experiments showed, and what it did not

Joaquin Philco (creator of the resonance architecture) · October 2026

Abstract

Joaquin Philco's resonance architecture puts banks of damped oscillators (resonators) on each axon, so a neuron reads the rhythm of its input as well as its level, and proposes that concepts form as structures of neurons that ring together. We tested the idea on vision, audio, letters, words and open-world novelty detection, always against controls that keep everything except the oscillation. The result is split. **As a sensor front end, resonance works**: on Spiking Heidelberg Digits an untrained resonator bank with a linear readout gets 85.7% against 82.7% for its low-pass control and 83.2% for the dataset paper's best network; on spoken words it matches a log-mel spectrogram and doubles low-pass; with eye movements it beats a static view of the same images and gives the best open-world detection we measured. **As a learning rule after the front end, it did not beat simple controls**: resonance-trained lattices tie frozen random ones, pruning during sleep never helped, and memory cells score the same with real, low-pass or phase-scrambled resonators.

1. The idea

Each connection is its own axon. The soma makes a time-domain decision (a sigmoid on a weighted sum). Along the axon, a sheath of resonators tuned to different frequencies reads the signal, $z_t = \rho\, e^{i\omega} z_{t-1} + x_t$, and reports amplitude $|z|$ and phase. A supporting cell (the oligodendrocyte in Joaquin's analogy) reads the resonators and adjusts the weight at the end of the axon. Learning is meant to happen by allocation: neurons in any layer that co-resonate on a new kind of input get linked into a structure for that concept, and a structure that always fires on the same input becomes long-term memory. The main goal throughout was semantic learning: a live model that meets something new (purple, an unknown word), recognises what kind of thing it is, and allocates a structure for it.

2. How we tested it

Every result below is compared with controls that remove only the resonance: **low-pass** (the same bank with $\omega = 0$, same decay), **phase-scrambled** resonators, **shuffled time order**, and simple learners on the same features (nearest class mean, a linear readout, Fuzzy ARTMAP). Runs used 3 seeds unless stated; later work showed 3-seed gains of a few points can be noise (standard error about 3 points), so we only claim gaps well above that.

3. Where resonance helps: the front end

Spiking Heidelberg Digits (SHD). 20 spoken digits as cochlear spike trains, test speakers partly unseen. A fixed resonator bank (3 to 48 Hz) on each pooled channel, read by a linear classifier, with no training of the front end.

Front end	Nearest mean	Linear (3 seeds)
Resonators	57.2	85.7
Low-pass control	52.2	82.7
Raw spike counts	49.6	75.0
Cramer et al. 2020, best RSNN		83.2
State of the art (trained)		96.3

Table 1. SHD test accuracy (%). The untrained bank beats its control and the dataset paper's best, about 10 points below trained state of the art.

Spoken words. 16 words from Speech Commands, 50 clips each, unseen speakers, the same small CNN on each input.

Ear	Nearest mean	Linear	CNN	5-way 1-shot / 5-shot
Resonator cochlea	20	45	73.5	35 / 48
Log-mel spectrogram	22	48	72.8	32 / 43
Low-pass control	9.5	12	36	27 / 33

Table 2. Spoken-word accuracy (%). Resonators match the standard spectrogram and double low-pass.

Eye movements. Images were never shown whole: a 12 x 12 fovea makes a stereotyped scan of fixations with small drift, and each input is read by resonators, including slow ones that span several fixations. On digits plus colours (18 classes, class-mean readout):

Input	Accuracy
Fovea + fast and slow resonators	92.6
Static full image	91.1
Fovea, resonator energy, fast only	83.7
Fovea, low-pass control	73.3
Fovea, shuffled fixation order	66.5

Table 3. The timing of the scan carries information that resonators pick up and low-pass filters lose.

Open-world novelty. Known classes are learned, then unknown classes arrive in the stream; each known class's sheet decides "is this mine?" and an image no sheet claims is flagged new.

Known classes, front end	Unknowns detected	False alarms	Known accuracy before / after
5, fovea resonators	82.5	12	97 / 94
5, fovea low-pass	74.6	10	87 / 81
15, fovea resonators	65	9	94 / 90
15, fovea low-pass	57	6	75 / 72
15, static Gabor features	36	7	88 / 84

Table 4. Resonators add about 8 points of detection and 13 to 18 points of known accuracy over low-pass, at 2 to 3 more false alarms.

On Fashion-MNIST (7 known, 3 unknown) the same scan scored a headline of 0.71 (harmonic mean of retention and detection) against 0.39 for low-pass, though a static Gabor class average still classified known items better (69.5 vs 65.8). Early on, a resonant sequence block also matched S4D on sequential MNIST (98.5 vs 98.3, one seed).

4. Where it did not: learning after the front end

Experiment	Resonance version	Control	Reading
3D lattice, resonance-trained links	83.5	84.1 (frozen random)	tie
Class-incremental agent vs Fuzzy ARTMAP (MNIST)	83.4	84.2	tie
Agent few-shot / novelty edge	present	same with class mean	prototype effect

Pruning during sleep	-0.8 to +0.2	no pruning	never helped
Memory cells, open world (detected)	75.4	75.4 low-pass / scrambled / frozen	tie
Memory cells vs class mean (accuracy)	78.6	90.3	worse

Table 5. Learning rules built on co-resonance did not beat frozen, prototype or low-pass controls.

The pattern is consistent. Wherever learning happened after the sensor, the useful work was done by prototypes (class averages of the input) and the resonant dynamics added nothing measurable. Lattices either ran away or stayed at the frozen reservoir's score, activity faded within a few slices, and freed space deep in a volume was never reached by new classes. Grouping by co-firing did find real structure (in a random volume, cells of the same ink linked 99% vs 18% by chance; error signals between two levels of cells found all six colours unsupervised), but class-average templates with the same step did about as well.

5. What we take from it

Resonance is a good way to *sense*: it turns timing (spike rhythms, speech modulation, the order of eye movements) into features that a simple readout can use, and it beats a fair low-pass control every time it sits at the sensor. It is not yet a learning rule: the co-resonance wiring, sleep and pruning rules we tried are matched by prototypes. The defensible claim today is **resonant sensors, with controls**. Open next steps are a learned resonant front end (V1 as a volume, learned the way active efficient coding learns it), Spiking Speech Commands, and judging any new learning rule on 9 or more runs against frozen and prototype controls.

Code and full results: the resonance-semantics project files (results_*.md for each experiment). Live models are in the other Resonance pages of this site. Idea and architecture: Joaquin Philco.